Principal Component Analysis and Machine Learning for Gym Experience Level Classification

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***Abstract*—Understanding and predicting exercise experience levels based on physiological and workout-related data can provide valuable insights for personalized fitness planning. This project investigates the application of Principal Component Analysis (PCA) and machine learning–based multiclass classification to analyze and predict gym members' experience levels using a real-world dataset.**

The Gym Members Exercise Tracking Dataset from Kaggle is used, containing body composition, demographic, and workout characteristics for 973 gym members. PCA is first applied to standardized numerical features to reduce dimensionality, remove redundancy among correlated variables, and reveal latent structure in the data. The PCA results show that three principal components capture over 92% of the total variance, providing an interpretable and compact representation of body size, body composition, and workout behavior.

Using this reduced feature space, supervised machine learning models are trained to classify gym members into Beginner, Intermediate, and Expert categories. Multiple classifiers are benchmarked using the PyCaret framework, and Random Forest and Naive Bayes are selected for detailed analysis. Random Forest achieves the best overall performance, particularly after hyperparameter tuning, while Naive Bayes serves as a competitive and interpretable baseline.

The results demonstrate that PCA effectively preserves discriminative information relevant to exercise experience, and ensemble-based classifiers are well suited for capturing nonlinear relationships in fitness-related data. This study highlights the value of combining dimensionality reduction with machine learning for practical fitness analytics applications.

Keywords—PCA, multiclass classification, PyCaret, Random Forest, fitness analytics

I. INTRODUCTION

With the increasing availability of fitness tracking and health-related data, data-driven methods have become an important tool for understanding exercise behavior

and physical conditioning. Modern gym datasets often include multiple physiological and workout-related variables, such as body composition measures and training duration, which together provide a rich description of an individual's fitness profile. However, analyzing such datasets presents challenges due to high dimensionality, correlated features, and nonlinear relationships among variables.

Machine learning techniques offer effective solutions for extracting meaningful patterns from complex datasets, while dimensionality reduction methods help simplify feature representations and improve interpretability. In particular, Principal Component Analysis (PCA) is widely used to reduce redundancy and visualize latent structure in multivariate data, and supervised classification algorithms can leverage these representations to make accurate predictions.

The objective of this project is to analyze and predict gym members' exercise experience levels using a combination of PCA and machine learning classification. The experience level is categorized into three classes—Beginner, Intermediate, and Expert—reflecting increasing levels of training proficiency and physical conditioning. First, PCA is applied to standardized numerical features to reduce dimensionality and explore dominant sources of variation. Then, machine learning classifiers are trained and evaluated to predict experience levels based on the available features.

The main contributions of this project are:

1. the application of PCA to reveal interpretable latent structure in gym-related data,
2. a systematic comparison of machine learning models for multiclass experience-level prediction, and
3. an evaluation of how dimensionality reduction supports classification performance and interpretability.

The remainder of this report is organized as follows. Section II introduces the PCA methodology, followed by the machine learning classification framework in Section III. Section IV describes the dataset, Section V presents the PCA results, and Section VI discusses the classification performance. Finally, Section VII summarizes the conclusions and outlines potential directions for future work.

II. PRINCIPAL COMPONENT ANALYSIS

Real-world datasets often exhibit high dimensionality, which increases computational cost and makes visualization and modeling more challenging. In addition, high-dimensional data frequently contain correlated or redundant variables that do not

provide independent information. Principal Component Analysis (PCA) is a widely used dimensionality reduction technique that addresses these challenges by transforming the original feature space into a smaller set of uncorrelated variables while preserving most of the variance in the data[2].

PCA reduces the complexity of a dataset by projecting the original variables onto a new coordinate system defined by principal components (PCs). Each principal component is a linear combination of the original variables and captures a specific amount of variance, with the first component explaining the largest variance and subsequent components explaining progressively less.

A. PCA Algorithm [1]

PCA can be applied to a data matrix X of dimension $n \times p$ through the following steps:

1. Standardization
   Since the original features may have different units and scales, the data are first standardized so that each variable contributes equally to the analysis. This is achieved by centering each feature around its mean and scaling it to unit variance.

2. Covariance Matrix Computation
   After standardization, a covariance matrix is computed to quantify the relationships among variables. Highly correlated variables indicate redundancy and motivate dimensionality reduction.

3. Eigen Decomposition
   Eigenvalues and eigenvectors of the covariance matrix are then computed. Each eigenvector represents the direction of a principal component, while the corresponding eigenvalue indicates the amount of variance captured by that component.

4. Projection onto Principal Components
   Finally, the standardized data are projected onto the eigenvector matrix to obtain the principal component scores. These transformed variables form a new feature space where the first few components retain most of the original information.

In this project, PCA was applied to the standardized numerical features of the gym dataset to reduce redundancy among correlated variables, facilitate visualization, and support subsequent classification. Both a manual PCA implementation and a PCA library–based implementation were explored, and consistent results were obtained. Due

to its flexibility and concise syntax, the PCA library implementation was primarily used for analysis and visualization.

III. MACHINE LEARNING–BASED CLASSIFICATION

After dimensionality reduction and exploratory analysis, supervised machine learning techniques were employed to classify gym members into three experience levels: Beginner, Intermediate, and Expert. This task is formulated as a multiclass classification problem.

Based on preliminary benchmarking using PyCaret, two representative and well-performing classifiers were selected for detailed analysis: Random Forest (RF) and Naive Bayes (NB). These models were chosen because they represent two fundamentally different classification paradigms: ensemble-based decision trees and probabilistic modeling.

A. Random Forest (RF)

Random Forest is an ensemble learning algorithm that constructs multiple decision trees using bootstrapped samples of the training data and randomly selected feature subsets[5]. The final prediction is obtained by aggregating the predictions of individual trees through majority voting.

The main advantage of Random Forest lies in its ability to model nonlinear decision boundaries and feature interactions, while reducing overfitting compared to a single decision tree. Random Forest is also relatively robust to noise and outliers, making it suitable for complex real-world datasets.

In this project, Random Forest was trained using stratified 10-fold cross-validation. Hyperparameter tuning was performed to improve generalization performance, and the tuned model demonstrated improved accuracy and F1-score compared to the baseline configuration.

B. Naive Bayes (NB)

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem, assuming conditional independence among features given the class label. Despite this simplifying assumption, Naive Bayes often performs well in practice, particularly when feature dimensionality is moderate and class distributions are reasonably separable.

The Gaussian Naive Bayes variant was used in this study, as the input features are continuous. Naive Bayes offers fast training, low computational cost, and strong interpretability, making it a useful baseline classifier.

Although Naive Bayes is less flexible than Random Forest in modeling complex nonlinear patterns, it provides competitive performance and serves as an effective point of comparison for evaluating the benefits of more sophisticated models.

IV. DATA SET DESCRIPTION

The dataset used in this project is the Gym Members Exercise Tracking Dataset, which was obtained from Kaggle[3]. This dataset contains physiological, demographic, and workout-related information for gym members and is designed to support the analysis of exercise behavior and experience levels. The primary objective of this dataset is to enable the prediction of a gym member's exercise experience level based on body composition and workout characteristics.

The target variable in the dataset is Experience Level, which represents the member's training background and is categorized into three classes:
Beginner (1), Intermediate (2), and Expert (3). These categories reflect increasing levels of workout proficiency and physical conditioning.

The original dataset consists of 973 records, where each record corresponds to one gym member profile. While the dataset contains a broad range of attributes, this study focuses on a subset of five continuous numerical features that describe body composition and workout duration, as these variables are most relevant for analyzing exercise experience. The selected features are:

- Weight (kg): body weight of the gym member
- Height (m): height of the gym member
- BMI: body mass index, a standardized measure of body composition
- Fat Percentage: proportion of body fat
- Session Duration (hours): length of each workout session

In addition, the dataset includes categorical attributes such as Gender and Workout Type, which are used in the classification stage but excluded from the PCA computation.

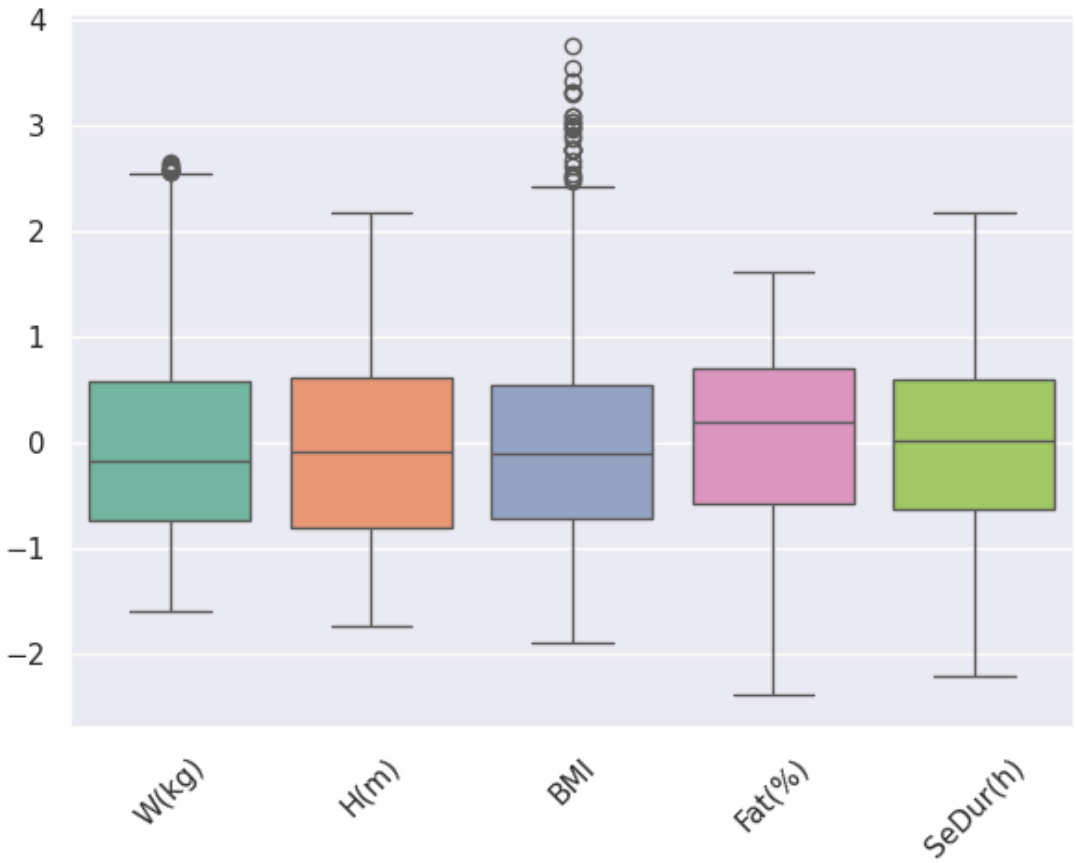

Fig. 1: Box plots of the five numerical features.

To explore the statistical properties of the dataset, box-and-whisker plots were used to examine the distributions, central tendencies, and variability of the selected features. Fig. 1 illustrates the box plots for the five numerical variables. It can be observed that most features exhibit approximately symmetric distributions, while noticeable outliers are present across all variables. In particular, BMI and Weight show larger variability and a wider spread, indicating significant differences in body composition among gym members. Session Duration exhibits a more compact distribution, reflecting typical workout time constraints.

Overall, this exploratory analysis highlights that the dataset contains both correlated and complementary features, making it suitable for dimensionality reduction using Principal Component Analysis (PCA). The presence of correlated variables motivates the application of PCA to reduce redundancy while preserving the dominant variance structure. These observations provide the foundation for the PCA analysis presented in the next section.

V. PCA RESULTS

Principal Component Analysis (PCA) is applied to the gym member exercise dataset in

order to reduce the dimensionality of the feature space and to analyze the dominant sources of variation among gym members. PCA was implemented using two approaches: (1) a manual implementation based on standard numerical operations to understand the underlying mathematical steps, and (2) a well-documented PCA library for efficient computation and visualization. Although both approaches produced consistent results, the PCA library implementation offers greater flexibility and cleaner visualization. Therefore, all figures and quantitative results presented in this section are obtained from the PCA library implementation.

Prior to applying PCA, exploratory data analysis was conducted to examine the statistical properties and relationships among the selected variables.

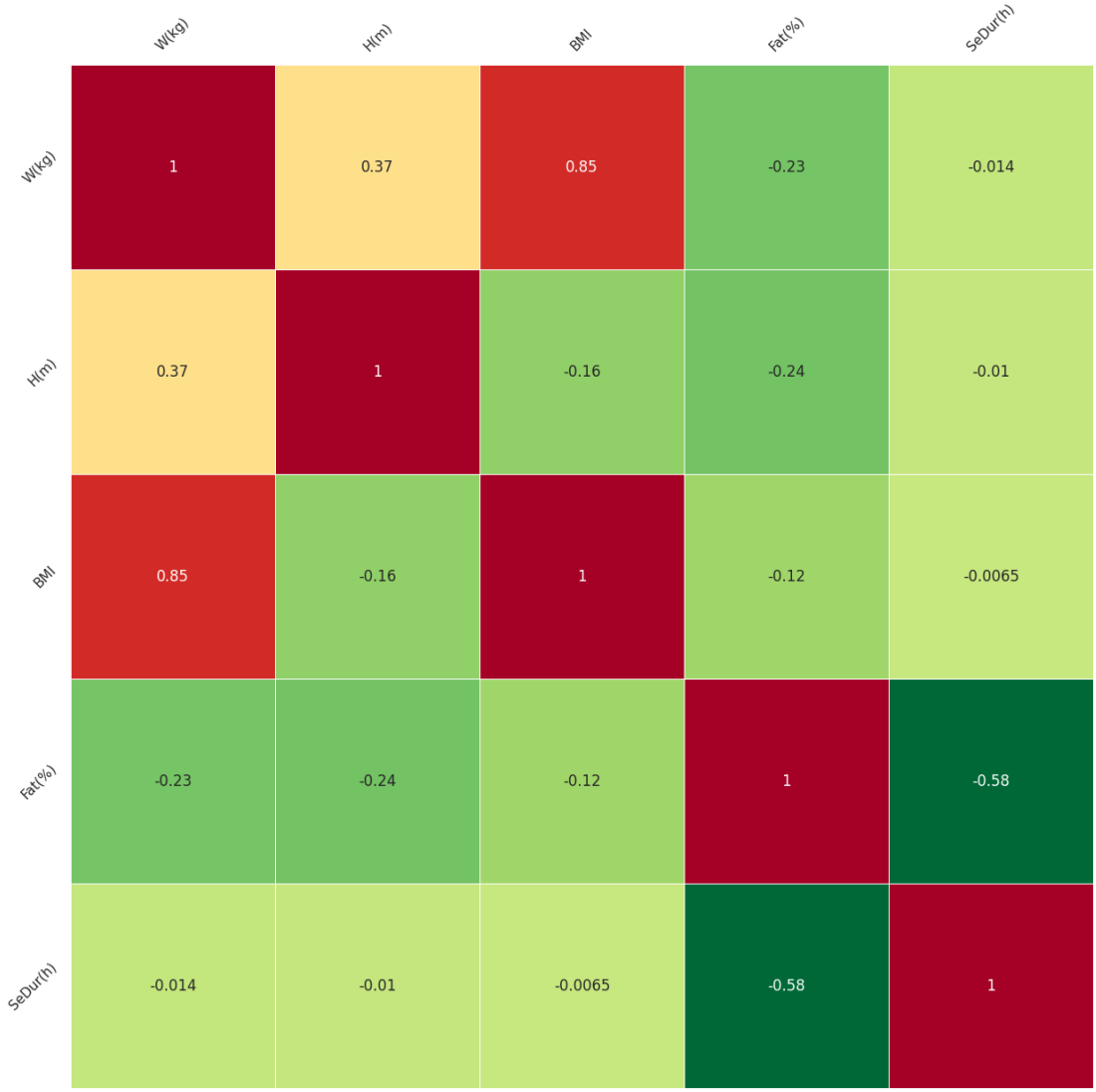

Fig. 2: Correlation matrix of the numerical features in the gym dataset.

Fig. 2 shows the correlation matrix of the five numerical features: weight, height, body mass index (BMI), fat percentage, and session duration. Strong positive correlation is

observed between weight and BMI, indicating redundancy in body composition information. In addition, fat percentage exhibits a moderate negative correlation with session duration, suggesting an inverse relationship between body fat and workout intensity or training commitment. These correlations motivate the application of PCA to obtain an uncorrelated and compact feature representation.

By applying the PCA transformation, the original $n \times p$dataset with $p = 5$features is reduced to $r$principal components, where $r < 5$. The dimensionality reduction is achieved through the eigenvector matrix $\mathbf{A}$, whose columns correspond to the principal components (PCs). Each principal component captures a portion of the total variance in the original dataset, and the number of retained components is determined based on their explained variance.

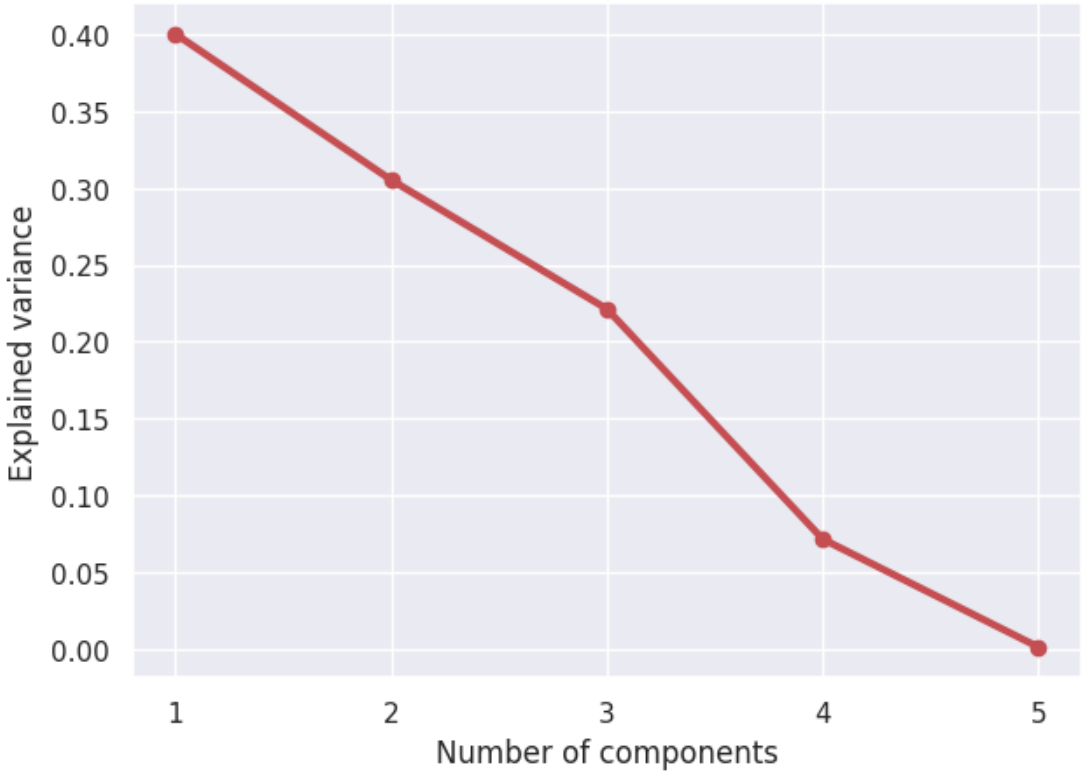

Fig. 3: Scree plot showing the explained variance ratio of each principal component.

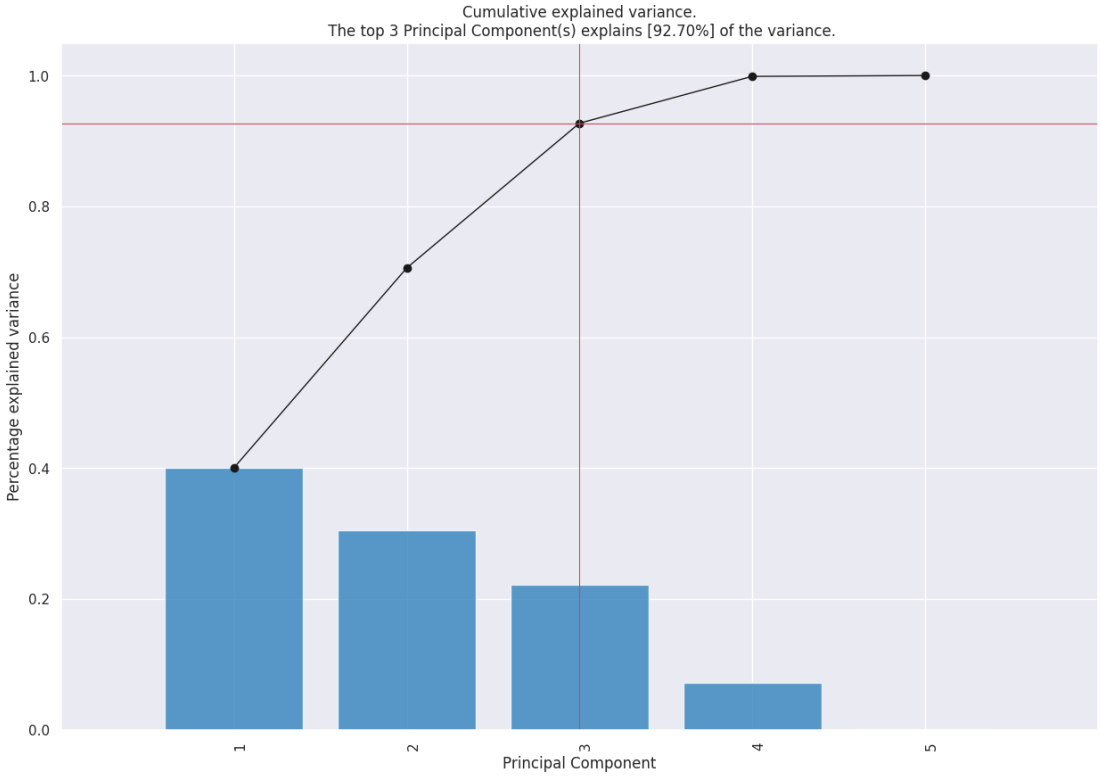

Fig. 4: Cumulative explained variance of principal components.

Fig. 3 and Fig. 4 illustrate the scree plot and the cumulative explained variance plot, respectively. The first principal component (PC1) explains approximately **40.0%** of the total variance, the second principal component (PC2) explains **30.5%**, and the third principal component (PC3) explains **22.2%**. Together, the top three principal components account for **92.7%** of the total variance in the dataset, while the remaining components contribute only marginally. The scree plot reveals a clear elbow after the third component, indicating that retaining three principal components provides an effective low-dimensional representation of the original feature space. Consequently, the dimensionality of the dataset is reduced from five to three for subsequent analysis.

The first principal component Z_1represents a body size–related dimension and is dominated by positive contributions from weight and BMI. This indicates that Z_1primarily captures variations in overall body mass and physique among gym members. The second principal component Z_2reflects a trade-off between body composition and workout behavior, with fat percentage contributing positively and session duration contributing negatively. This component distinguishes individuals with higher body fat and shorter workout sessions from those with lower body fat and longer training duration. Height contributes moderately to both principal components and plays a supporting role in describing overall body structure. The third principal component captures additional variance that complements the first two components,

but its physical interpretation is less pronounced and therefore not emphasized.

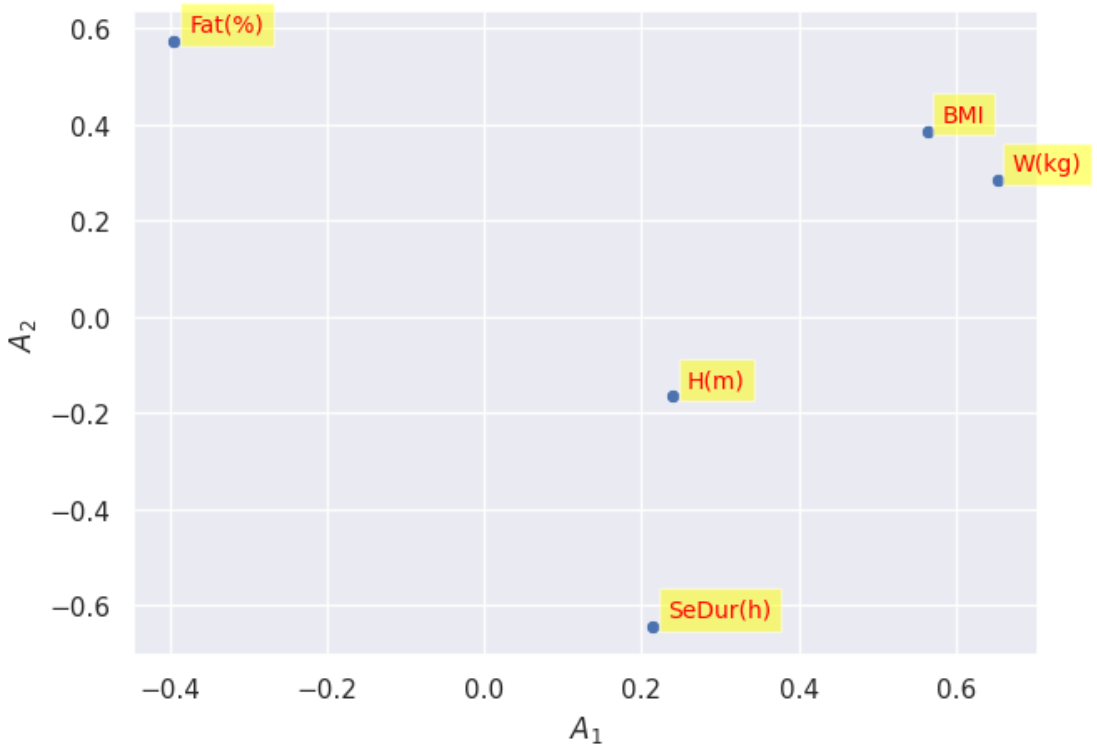

Fig. 5: Eigenvectors Component Coefficient Plot of principal components.

Fig. 5 presents the eigenvectors component coefficient plot, which visually summarizes the contributions of each original variable to the first two principal components. The plot confirms that session duration has the highest contributions to PC1, while fat percentage dominates PC2 with opposite signs. These observations are consistent with the numerical loading values obtained from the PCA library implementation.

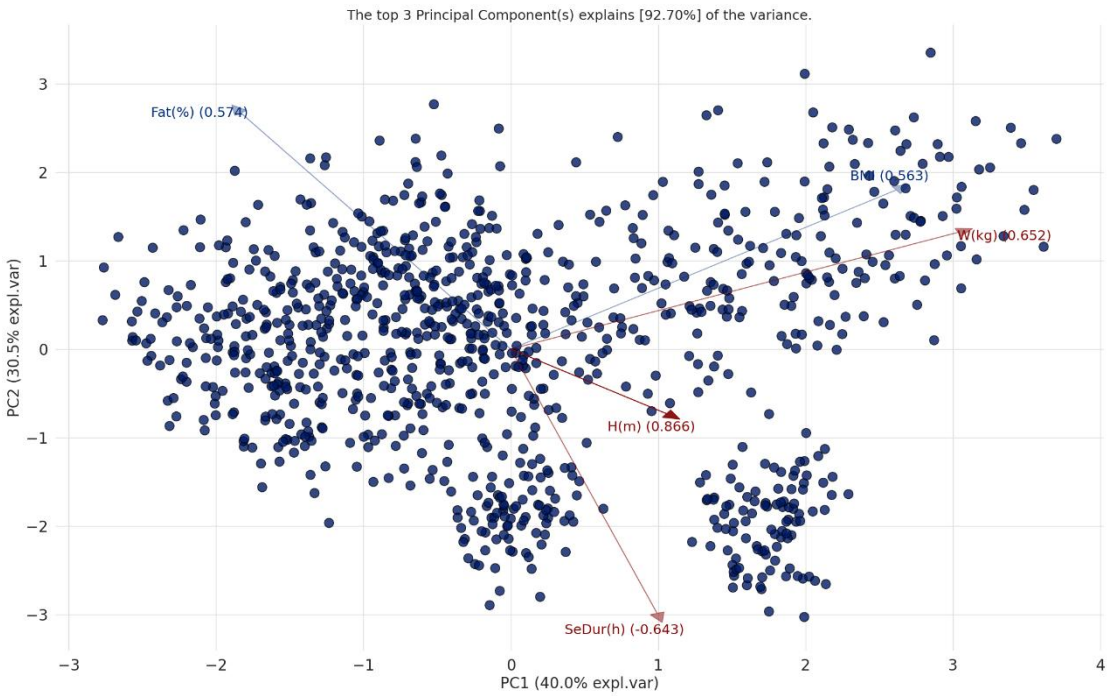

Fig. 6: PCA biplot showing observations and variable loadings in the PC1–PC2 space.

The biplot shown in Fig. 6 provides a combined visualization of both observations and variable loadings in the PC1–PC2 space. Each observation is represented as a point, while each original feature is shown as a vector originating from the origin. Features pointing in similar directions, such as fat percentage, are positively correlated, whereas features pointing in opposite directions, such as session duration, exhibit negative correlation. The relative angles and vector lengths further support the interpretation of the principal components discussed above.

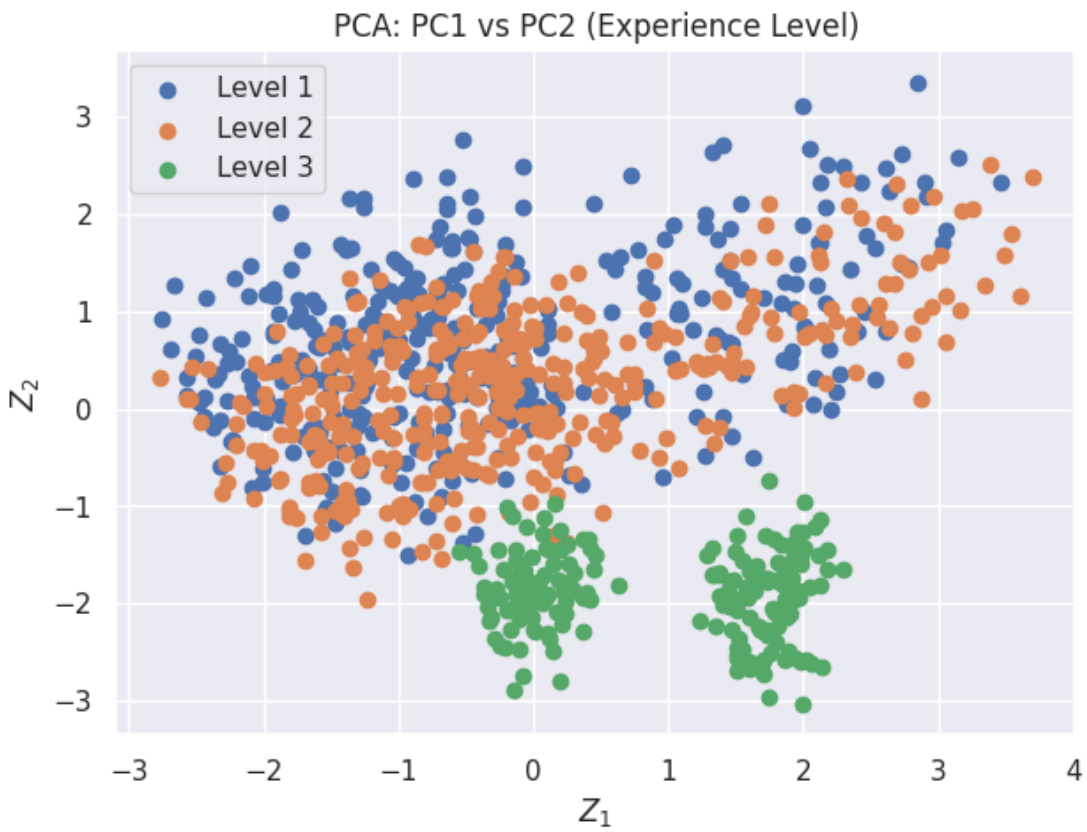

Fig. 7: Projection of gym members onto the first two principal components (PC1 vs PC2) colored by experience level.

Finally, Fig. 7 shows the projection of the dataset onto the first two principal components. Although three principal components are retained to preserve over 90% of the variance, the PC1–PC2 projection already reveals meaningful structural patterns. Different experience levels exhibit distinguishable distribution trends in the reduced-dimensional space. In particular, advanced-level gym members tend to cluster in regions associated with lower fat percentage and longer session duration, while beginner-level members are more concentrated in regions characterized by higher fat percentage and shorter workouts. This indicates that the principal components preserve discriminative information relevant to exercise experience level and provide a suitable low-dimensional representation for subsequent multiclass classification.

Overall, PCA effectively reduces the dimensionality of the original feature space while retaining the majority of variance and revealing interpretable latent structure. The resulting principal components serve as informative and compact inputs for the classification models discussed in the next section.

VI. CLASSIFICATION RESULTS

This section presents the multiclass classification results for predicting Experience

Level (1 = Beginner, 2 = Intermediate, 3 = Expert) using the selected physiological and workout-related features. The modeling pipeline was implemented using PyCaret (classification module)[4]. As introduced earlier, 10% of the original dataset was held out as unseen data to simulate a real-world deployment scenario, while the remaining 90% (876 samples) was used for model development. Within the modeling set, PyCaret performed an internal train/test split (70%/30%) and evaluated models using Stratified 10-fold cross-validation, ensuring robust performance estimates under class-balanced folds.

Model Benchmarking and Best Model Selection

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
rf	Random Forest Classifier	0.7583	0.8474	0.7583	0.7675	0.7497	0.6132	0.6273	0.8800
xgboost	Extreme Gradient Boosting	0.7519	0.8491	0.7519	0.7544	0.7489	0.6048	0.6094	0.2220
nb	Naive Bayes	0.7503	0.8320	0.7503	0.7627	0.7428	0.6010	0.6155	0.5100
et	Extra Trees Classifier	0.7403	0.8418	0.7403	0.7415	0.7370	0.5867	0.5909	0.3820
lightgbm	Light Gradient Boosting Machine	0.7389	0.8436	0.7389	0.7416	0.7350	0.5838	0.5895	0.3610
lr	Logistic Regression	0.7373	0.0000	0.7373	0.7426	0.7328	0.5818	0.5895	2.6910
lda	Linear Discriminant Analysis	0.7372	0.0000	0.7372	0.7418	0.7321	0.5812	0.5891	0.1290
gbc	Gradient Boosting Classifier	0.7370	0.0000	0.7370	0.7412	0.7301	0.5799	0.5895	0.7190
ridge	Ridge Classifier	0.7144	0.0000	0.7144	0.7153	0.7086	0.5465	0.5529	0.4570
dt	Decision Tree Classifier	0.6963	0.7436	0.6963	0.6993	0.6957	0.5197	0.5218	0.2900
ada	Ada Boost Classifier	0.6804	0.0000	0.6804	0.7093	0.6608	0.4977	0.5306	0.4200
qda	Quadratic Discriminant Analysis	0.6675	0.0000	0.6675	0.6745	0.6618	0.4717	0.4813	0.1980
knn	K Neighbors Classifier	0.6069	0.7350	0.6069	0.6059	0.6053	0.3770	0.3780	0.3260
svm	SVM - Linear Kernel	0.5582	0.0000	0.5582	0.4823	0.4592	0.2904	0.3694	0.2740
dummy	Dummy Classifier	0.4356	0.5000	0.4356	0.1898	0.2643	0.0000	0.0000	0.0760

Fig. 8: Benchmark comparison of machine learning models using stratified 10-fold cross-validation

To identify the most suitable classifier, multiple machine learning models were benchmarked using PyCaret's compare_models() function. The evaluation metrics include Accuracy, Recall, Precision, F1-score, Kappa, and MCC (Matthews Correlation Coefficient). The results indicate that the top three models are:

- Random Forest (rf): Accuracy ≈ 0.7583, F1 ≈ 0.7497
- Extreme Gradient Boosting (xgboost): Accuracy ≈ 0.7519, F1 ≈ 0.7489
- Naive Bayes (nb): Accuracy ≈ 0.7503, F1 ≈ 0.7428

Although the three models achieved similar baseline performance, Random Forest obtained the highest overall accuracy and F1-score and was selected as the primary candidate for further tuning and evaluation. For completeness and comparative analysis, Naive Bayes were also trained and analyzed using the same workflow.

Random Forest (RF) – Baseline and Tuned Model

Fold	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	0.7742	0.8380	0.7742	0.7906	0.7662	0.6374	0.6554
1	0.8710	0.9215	0.8710	0.8721	0.8705	0.7944	0.7957
2	0.8065	0.9097	0.8065	0.8102	0.8043	0.6908	0.6954
3	0.7377	0.8148	0.7377	0.7512	0.7290	0.5815	0.5983
4	0.7049	0.8222	0.7049	0.7289	0.6805	0.5300	0.5667
5	0.7541	0.8675	0.7541	0.7537	0.7537	0.6138	0.6140
6	0.7049	0.7801	0.7049	0.7099	0.6929	0.5247	0.5401
7	0.7377	0.7899	0.7377	0.7477	0.7270	0.5775	0.5945
8	0.7377	0.8760	0.7377	0.7477	0.7270	0.5775	0.5945
9	0.7541	0.8542	0.7541	0.7625	0.7461	0.6046	0.6180
Mean	0.7583	0.8474	0.7583	0.7675	0.7497	0.6132	0.6273
Std	0.0472	0.0450	0.0472	0.0440	0.0523	0.0760	0.0695

Fig. 9: Cross-validation performance of the baseline Random Forest classifier

The Random Forest model was trained using PyCaret's create_model('rf') and evaluated via stratified 10-fold cross-validation. The baseline RF achieved:

- Mean Accuracy = 0.7583
- Mean F1 = 0.7497
- Mean Precision = 0.7675

The standard deviation across folds (Std Accuracy = 0.0472; Std F1 = 0.0523) suggests moderate variability and relatively stable generalization.

Fold	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	0.7742	0.8393	0.7742	0.8401	0.7561	0.6376	0.6845
1	0.8710	0.9044	0.8710	0.9005	0.8664	0.7928	0.8152
2	0.8548	0.8924	0.8548	0.8911	0.8485	0.7666	0.7944
3	0.7869	0.8460	0.7869	0.8579	0.7686	0.6585	0.7131
4	0.7213	0.8011	0.7213	0.8315	0.6740	0.5532	0.6409
5	0.8525	0.8947	0.8525	0.8904	0.8454	0.7659	0.7942
6	0.7541	0.8150	0.7541	0.8419	0.7213	0.5990	0.6710
7	0.7705	0.8273	0.7705	0.8489	0.7439	0.6264	0.6902
8	0.7541	0.8150	0.7541	0.8419	0.7213	0.5990	0.6710
9	0.8197	0.8643	0.8197	0.8719	0.8066	0.7080	0.7495
Mean	0.7959	0.8500	0.7959	0.8616	0.7752	0.6707	0.7224
Std	0.0481	0.0353	0.0481	0.0237	0.0609	0.0786	0.0584

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Fitting 10 folds for each of 10 candidates, totalling 100 fits
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Fig. 10: Cross-validation performance of the tuned Random Forest classifier

To improve performance, hyperparameter tuning was performed using tune_model(), which applies randomized grid search over a predefined hyperparameter space and evaluates candidates with stratified 10-fold CV. After tuning, RF improved notably:

- Mean Accuracy = 0.7959 (↑ from 0.7583)
- Mean F1 = 0.7752 (↑ from 0.7497)
- Mean Precision = 0.8616(↑ from 0.7675)

The tuned RF adopts a more regularized configuration (e.g., reduced tree depth and stricter minimum samples per split/leaf), which reduces overfitting and improves generalization.

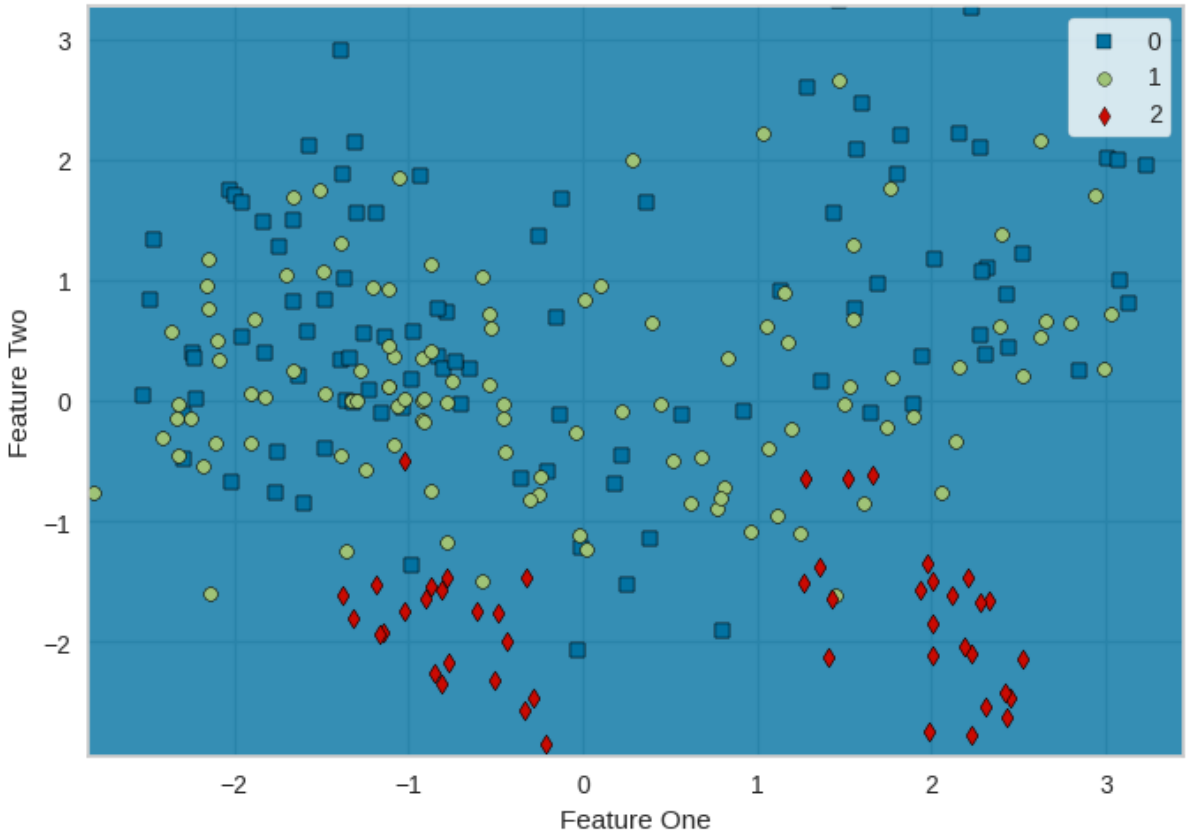

Fig. 11: Decision boundary visualization of the Random Forest classifier in reduced feature space.

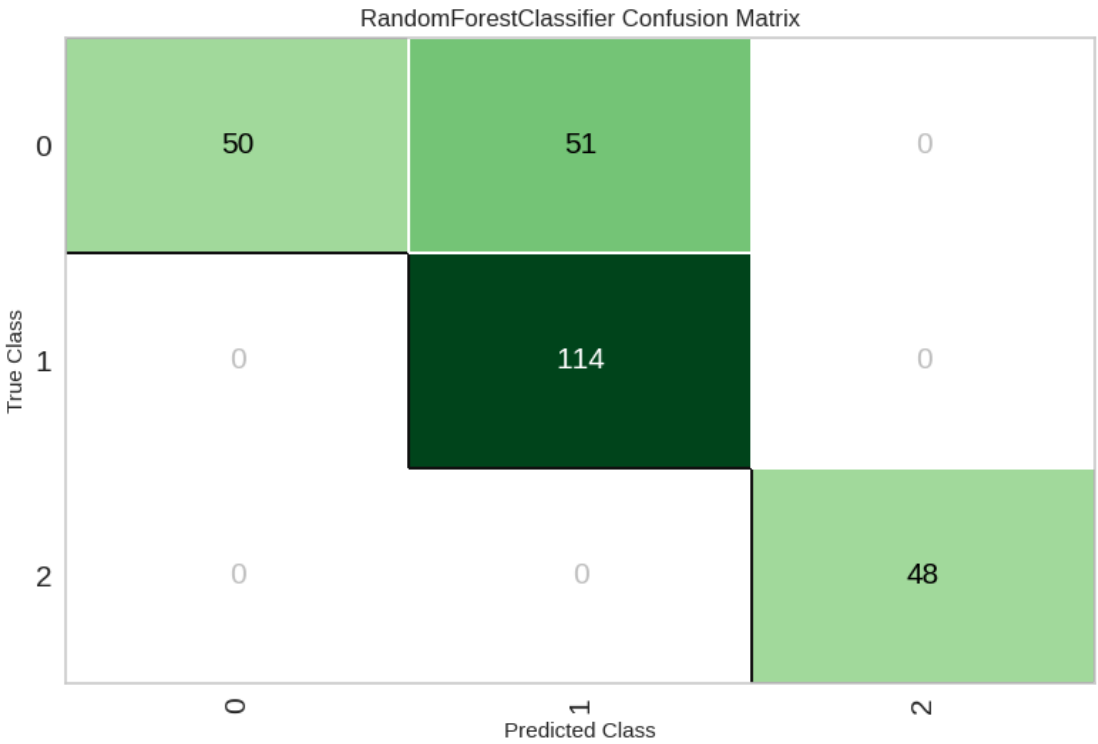

Fig. 12: Confusion matrix of the Random Forest classifier.

Fig. 11 shows the decision boundary visualization (2D projection) for Random Forest, illustrating how the model separates the three classes in a nonlinear manner. Fig. 12 presents the confusion matrix of the RF model. The results indicate that the model performs particularly well on one class (with 114 correct predictions), while the primary error pattern occurs between Beginner vs Intermediate, reflecting a realistic overlap between these two experience levels in terms of body composition and workout duration patterns. The Expert class (Class 2) demonstrates strong separability with almost all samples correctly classified (48 correct), which is consistent with the PCA findings where advanced members form clearer clusters in the reduced space.

Naive Bayes (NB) – Baseline Model and Tuning Outcome

Fold	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	0.7097	0.8194	0.7097	0.7100	0.7065	0.5362	0.5398
1	0.7903	0.8818	0.7903	0.7911	0.7905	0.6671	0.6674
2	0.8065	0.8818	0.8065	0.8167	0.8023	0.6900	0.7007
3	0.7541	0.8341	0.7541	0.7752	0.7439	0.6073	0.6298
4	0.6721	0.7832	0.6721	0.6737	0.6595	0.4804	0.4942
5	0.8525	0.8866	0.8525	0.8904	0.8454	0.7659	0.7942
6	0.6885	0.7387	0.6885	0.6932	0.6729	0.4974	0.5162
7	0.7377	0.8269	0.7377	0.7377	0.7377	0.5833	0.5833
8	0.7705	0.8260	0.7705	0.8182	0.7509	0.6277	0.6725
9	0.7213	0.8410	0.7213	0.7205	0.7189	0.5549	0.5571
Mean	0.7503	0.8320	0.7503	0.7627	0.7428	0.6010	0.6155
Std	0.0531	0.0440	0.0531	0.0642	0.0549	0.0849	0.0895

Fig. 13: Cross-validation performance of the baseline Naive Bayes classifier

Naive Bayes was trained using create_model('nb'). Despite its simplifying conditional-independence assumption, NB performed competitively:

- Mean Accuracy = 0.7503

- Mean F1 = 0.7428
- Mean Precision = 0.7627

Hyperparameter tuning was attempted, but PyCaret reported that the original model performed better than the tuned model, and therefore the baseline model was retained. This result is expected because Naive Bayes has a limited hyperparameter space and often does not benefit substantially from tuning compared with ensemble tree-based methods.

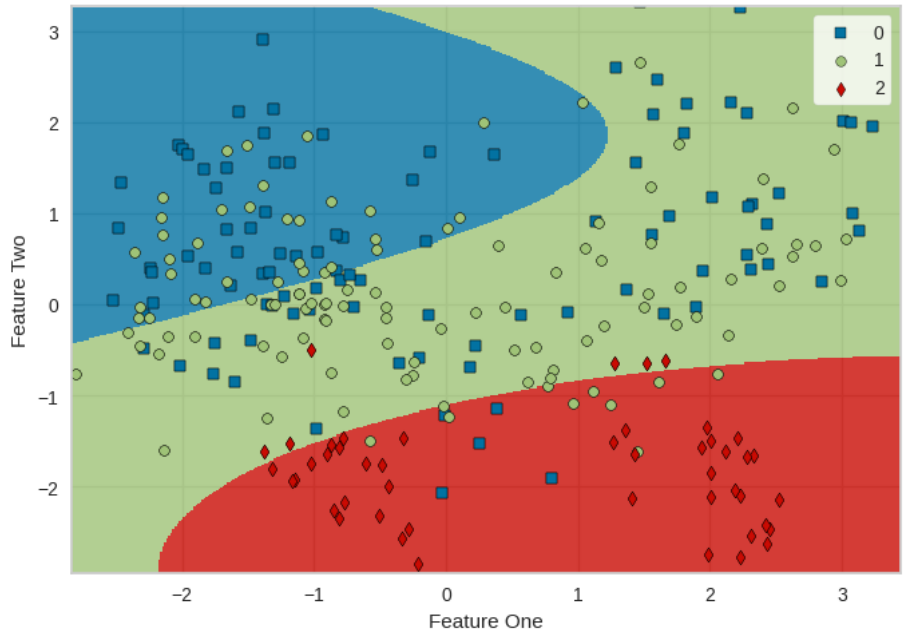

Fig. 14: Decision boundary visualization of the Naive Bayes classifier in reduced feature space

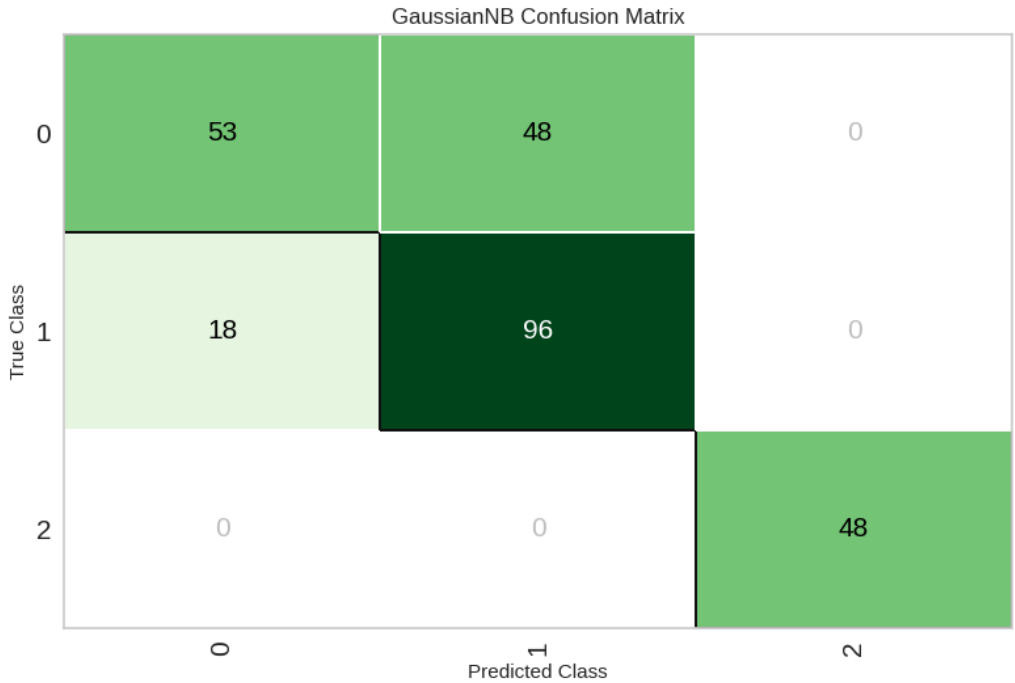

Fig. 15: Confusion matrix of the Naive Bayes classifier

Fig. 14 shows the Naive Bayes decision boundary plot, which reflects the model's relatively simple separation strategy. Fig. 15 presents the confusion matrix for NB.

Compared with RF and XGBoost, NB shows increased confusion between Beginner and Intermediate classes, and also introduces additional misclassifications for Intermediate samples (e.g., intermediate predicted as beginner). This indicates that NB is less capable of capturing the nonlinear relationships among body composition variables and workout duration. Nevertheless, NB remains a strong baseline due to its simplicity, fast training, and reasonably high accuracy.

Summary of Classification Findings

Across all evaluated models, ensemble tree-based methods (Random Forest) achieved the best performance, consistent with the dataset characteristics where nonlinear interactions exist among body measurements and workout duration. Hyperparameter tuning produced meaningful improvements for both RF and XGBoost, with tuned accuracy approaching 0.80 and improved F1-scores. The major classification difficulty consistently arises from overlap between Beginner and Intermediate experience levels, while the Expert group is more separable and consistently classified with high accuracy. These findings align with the PCA visualization results, where the advanced group forms clearer clusters in the principal component space.

In the next section, the final conclusions are summarized and potential improvements are discussed.

VII. CONCLUSION AND FUTURE WORK

This project explored the integration of Principal Component Analysis (PCA) and machine learning–based multiclass classification to analyze and predict gym members' exercise experience levels using physiological and workout-related data. PCA was successfully applied to reduce the dimensionality of the original feature space while preserving the majority of variance and revealing meaningful latent structure related to body size, body composition, and workout behavior.

The PCA results showed that three principal components captured over 92% of the total variance, providing a compact and interpretable representation of the data. Visualization in the principal component space demonstrated that gym members with higher experience levels tend to cluster in regions associated with lower fat percentage and longer workout duration, indicating that the reduced feature space retains discriminative information relevant to experience level.

For classification, multiple machine learning models were evaluated using the PyCaret framework. Among them, Random Forest achieved the best overall performance, particularly after hyperparameter tuning, with improved accuracy and F1-score. Naive Bayes, while less flexible, provided competitive performance and served as a strong and interpretable baseline. Across models, the main classification challenge was distinguishing between Beginner and Intermediate members, whereas the Expert class exhibited clearer separability, consistent with the PCA findings.

Overall, the results demonstrate that combining PCA with ensemble-based classifiers is effective for fitness-related classification tasks involving correlated and nonlinear features. The proposed workflow provides a practical approach for analyzing gym data and predicting exercise experience levels.

Future work could explore several extensions. Additional features such as training frequency, heart rate metrics, or longitudinal workout data could be incorporated to improve predictive performance. Alternative dimensionality reduction techniques and advanced classifiers, such as neural networks, could also be investigated. Finally, deploying the trained models in a real-time fitness recommendation system would be a valuable direction for practical application.

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